

ACRME Complete Requirements Reference

This document consolidates **every requirement for the Azure Capacity Reservation Management Engine (ACRME)**, spanning Functional Requirements (FR), Non-Functional Requirements (NFR), and Placement-Specific Requirements (R), organized by category with evidence of coverage and design maturity.

Sources: - azure_cr_management_engine_design.md (FR-1..8, NFR-1..7)
- acrme_production_readiness_review_and_architecture.md (Section 22 Must/Should/Could; cross-references to PRR) - acrme_requirements_traceability_review.md (coverage matrix + blockers) - multi_region_placement_design.md (Placement Requirements R1..R8, NFR-R1..R3)

Classification: Principal Cloud Architect — System Design

Maturity: Production Readiness Review Complete; POC-gated execution ready

Part 1 — Functional Requirements (FR-1 through FR-8)

FR-1 — Capacity Reservation Lifecycle Management

Manages the full CRUD lifecycle of Capacity Reservation Groups (CRGs) and Capacity Reservations (CRs) across the Azure estate at **api-version=2024-03-01** or later.

ID	Requirement	Coverage	Implementation
FR-1.1	CRG create, read, update, delete via Azure Compute REST API at API version 2024-03-01+	COVERED	EPIC-02 (Provisioning Engine); API endpoints: POST/GET/PATCH/DELETE /api/v1/crgs
FR-1.2	CR create, read, update, delete within managed CRGs; support quantity modification without VM disruption	COVERED	EPIC-02 stories E02-S03, E02-S04; PATCH /crgs/{id}/reservations/{cr_id} accepts new quantity

ID	Requirement	Coverage	Implementation
FR-1.3	Zero-size reservation pattern: create CRs at quantity=0 , increment as VMs associate	COVERED	E02-S07; CapacityReservation.desired_quantity permits 0; P1 backlog item
FR-1.4	Track allocated (VMs currently associated) vs quantity (reserved capacity) independently; alert on overallocation	COVERED	CapacityReservation entity; E05-S06 (Overallocation Detection); GET .../utilization API
FR-1.5	Enforce CR deletion sequencing: disassociate all VMs → delete all CRs → delete CRG; reject out-of-order deletes	COVERED	E02-S05, E02-S06; cascade validation guards in Provisioning Service
FR-1.6	Support CR quantity reduction; enforce floor constraint (quantity allocated)	COVERED	E02-S04; floor validation guard prevents reduction below allocated count

FR-1 Verdict: 6/6 covered — Fully Implemented

FR-2 — Capacity Reservation Sharing Management

Orchestrates the three-step RBAC setup for Provider-Consumer CRG sharing, managing subscription membership and enforcing constraints.

ID	Requirement	Coverage	Implementation
FR-2.1	Manage <code>sharingProfile.subscriptionIds</code> : add, remove, list Consumer subscriptions on CRGs	COVERED	EPIC-04 (Sharing Management); POST/GET/DELETE <code>/crgs/{id}/consumers</code> ; updates ARM <code>sharingProfile</code>
FR-2.2	Orchestrate complete three-step RBAC setup for each Provider- Consumer pair: (1) grant Provider <code>deploy/action</code> on Consumer sub, (2) add Consumer to CRG <code>sharingProfile</code> , (3) grant Consumer identity read + <code>deploy/action</code> on CRG	COVERED	E04-S02, E04-S06 (RBAC Orchestration); three-step sequence in <code>SharingService</code> ; atomic guard
FR-2.3	Enforce 100-Consumer- per-CRG limit; return structured error with remediation when limit reached	COVERED	E04-S01 (Constraint Validation); <code>CRG.max_consumer_count</code> = 100; HC-2 hard constraint
FR-2.4	Verify <code>Microsoft.Compute</code> provider registration in all Consumer subscriptions before completing sharing setup	COVERED	E04-S07 (Provider Registration Validation); pre-gate in onboarding workflow

ID	Requirement	Coverage	Implementation
FR-2.5	Support safe unsharing: detect active Consumer VM associations before removal; require explicit force-override	COVERED	E04-S03 (Safe Unsharing); check allocated count before removal; warn on forced unsharing
FR-2.6	Warn operators when forced unsharing is requested with active Consumer VMs citing silent hazard	COVERED	E04-S03; alert <code>ForceUnsharingWithActiveVMs</code> (Warning severity)
FR-2.7	Maintain current and historical sharing profile state per CRG; record timestamps of add/remove operations	COVERED	SharingRelationship entity; audit log; E04-S09 (History Tracking)

FR-2 Verdict: 7/7 covered — Fully Implemented

FR-3 — Availability Zone Mapping Management

Discovers, stores, and resolves logical-to-physical zone mappings across Provider-Consumer subscription pairs.

ID	Requirement	Coverage	Implementation
FR-3.1	Discover and store logical-to-physical zone mapping for every managed subscription in every managed region	COVERED	EPIC-09 (Zone Management); <code>ZoneMappingService</code> ; E09-S01 (Discovery)

ID	Requirement	Coverage	Implementation
FR-3.2	Use Check Zone Peers API to compute cross-subscription zone equivalence for all Provider-Consumer pairs	COVERED	E09-S02 (Cross-Subscription Equivalence); stores in zone mapping registry
FR-3.3	Maintain zone mapping registry: (provider_sub_id, region, provider_zone) → [(consumer_sub_id, consumer_zone)]	COVERED	ZoneMappingRecord entity (Cosmos DB); E09-S01
FR-3.4	Automatically resolve correct Consumer logical zone when accepting VM deployment against shared CRG	COVERED	E09-S03 (Zone Resolution); used in placement decision tree
FR-3.5	Detect and reject VM deployment where Consumer zone does not map to same physical zone as target CR	COVERED	E09-S04 (Zone Mismatch Rejection); hard constraint HC-8 (ZONE_ALIGNMENT)
FR-3.6	Refresh zone mappings on configurable schedule and on new subscription onboarding event	COVERED	E09-S05 (Scheduled Refresh); refresh cadence in PlacementPolicy; triggered on onboarding

FR-3 Verdict: 6/6 covered — Fully Implemented

FR-4 — Quota Management

Queries, tracks, and enforces quota constraints across managed subscriptions and quota groups.

ID	Requirement	Coverage	Implementation
FR-4.1	Query and store regional compute quota (<code>currentValue</code> , <code>limit</code>) for all tracked VM SKUs	COVERED	EPIC-03 (Quota Management); E03-S01 (Quota Discovery); QuotaRecord entity
FR-4.2	Calculate and expose derived quota metrics: $\text{available} = \text{limit} - \text{current}$; $\text{committed} = \Sigma(\text{CR qty} \times \text{vCPU})$; headroom; overcommit flag	COVERED	E03-S02 (Quota Metrics); RegionalSnapshot.quota_fields; GET <code>/quota/{region}</code> API
FR-4.3	Detect quota exhaustion before CR creation; block and return pre-validation failure with remediation guidance	COVERED	E03-S05 (Pre-Validation); hard constraint HC-3 (QUOTA_FLOOR); placement gate
FR-4.4	Initiate Azure quota increase requests via Support REST API when thresholds breached; subject to operator approval	COVERED	E03-S06 (Quota Increase Workflow); operator approval gate in Phase 1

ID	Requirement	Coverage	Implementation
FR-4.5	Track Consumer quota independently; warn when Consumer available quota insufficient for implied VM deployment	COVERED	E03-S05 (Consumer Quota Validation); quota pre-check before placement recommendation
FR-4.6	Unified quota dashboard across all managed subscriptions: aggregate reserved, consumed, available per region, zone, SKU	COVERED	EPIC-11 (Observability); E11-S02 (Grafana Dashboards); GET /dashboards/quota-overview

FR-4 Verdict: 6/6 covered — Fully Implemented

FR-5 — Disaster Recovery Failover Management

Manages pre-positioned DR capacity pairs, failover triggering, and failback orchestration.

ID	Requirement	Coverage	Implementation
FR-5.1	Support definition of DR capacity pairs: (primary_crg, dr_crg) where DR CRG is pre-positioned shared reserve	COVERED	EPIC-06 (DR Management); DRCapacityPair entity; E06-S01 (Pair Definition)

ID	Requirement	Coverage	Implementation
FR-5.2	Manage DR CRG sharing profiles; ensure DR Consumer has correct RBAC and zone mapping at all times	COVERED	E06-S02 (DR Pair Setup); pre-positions sharing before failover event
FR-5.3	Support failover trigger: validate DR CRG capacity, initiate bulk VM deployment from DR Consumer	COVERED	E06-S03 (Failover Trigger); pre-gated by state machine (EngineModeState); requires DR_EVENT_ACTIVE mode
FR-5.4	Support failback trigger: deallocate DR VMs, restore primary CRG to pre-failover config, record failback metadata	COVERED	E06-S04 (Failback Trigger); state transition FAILBACK_PENDING → STEADY_STATE
FR-5.5	Monitor DR CRG capacity continuously; alert if DR reserved capacity consumed in steady-state	COVERED	E05-S01 (Reconciliation Engine); alert UnauthorizedDRConsumption (Critical)
FR-5.6	Enforce minimum DR capacity buffers per DR pair as percentage of primary; alert when quota changes would violate	COVERED	E03-S11 (DR Floor Enforcement); HC-6, HC-7; dr_ratio_min=0.30, dr_ratio_max=0.40

ID	Requirement	Coverage	Implementation
FR-5.7	Support cross-region DR pair definitions with separate CRGs per region, independent sharing profiles and zone mappings	COVERED	E06-S01, E06-S05 (Cross-Region DR); Middle East cross-geo extension

FR-5 Verdict: 7/7 covered — Fully Implemented

FR-6 — Regional Placement Decisions

Evaluates VM placement against multi-dimensional constraints and returns ranked recommendations.

ID	Requirement	Coverage	Implementation
FR-6.1	Evaluate placement against: AZ physical alignment, CR capacity, Consumer quota headroom, SKU availability, cost, DR buffer compliance	COVERED	EPIC-08 (Placement Engine); E08-S01 through E08-S04; placement decision tree
FR-6.2	Return ranked list of valid placements when multiple CRGs/regions satisfy constraints, ordered by configurable policy	COVERED	E08-S05 (Policy-Driven Ranking); $\text{argmax}(\text{PS_Prod})$ / $\text{argmax}(\text{PS_NonProd})$ / $\text{argmax}(\text{PS_DR})$ scoring

ID	Requirement	Coverage	Implementation
FR-6.3	Enforce placement policies as code: operators define placement rules as structured policy documents; engine evaluates against policy DAG	COVERED	PlacementPolicy entity; version-controlled config
FR-6.4	Detect single-zone dependency (all VMs in same physical zone) and warn operator	COVERED	E08-S06 (Zone Diversity Warning); alert SingleZoneDependency (Warning)
FR-6.5	Surface SKU availability constraints per region and zone; integrate Azure Compute SKU API	COVERED	E08-S03 (SKU Availability); RegionalSnapshot.sku_availability_flags

FR-6 Verdict: 5/5 covered — Fully Implemented

FR-7 — Capacity Forecasting

Analyzes historical allocation patterns and produces demand forecasts with sizing recommendations.

ID	Requirement	Coverage	Implementation
FR-7.1	Analyze historical CR allocation patterns; produce demand forecasts for configurable period (30/60/90 days)	COVERED	EPIC-09 (Forecasting); E09-S01 (Historical Analysis)

ID	Requirement	Coverage	Implementation
FR-7.2	Produce capacity recommendations: increase CR qty when forecast > current, decrease when forecast consistently below	COVERED	E09-S02 (Recommendations); feeds into approval-gated auto-increase
FR-7.3	Model capacity buffer: <code>recommended_qty = ceil(forecast_peak × (1 + growth_buffer) + dr_buffer)</code>	COVERED	Forecast_Quantity formula
FR-7.4	Alert when forecast demand approaches quota limits (default 80% threshold) with 14-day lead time	COVERED	E09-S03 (Quota Alert); <code>ForecastApproachingQuotaLimit</code> alert
FR-7.5	Support workload tagging: associate VMs and CRs with named workloads; enable per-workload capacity forecasts	PARTIALLY COVERED	E09-S04 (Workload Tagging); implementation detail incomplete
FR-7.6	Expose raw forecast data (time series) and derived recommendations via API for external capacity planning tools	COVERED	<code>GET /forecasts/{crg_id}</code> API; returns <code>forecast_peak</code> , <code>recommendations</code> , <code>time_series</code>

FR-7 Verdict: 5/6 covered; 1 partially — Workload tagging implementation detail deferred

FR-8 — Cost Optimization

Monitors utilization, calculates costs, and recommends right-sizing and charge-back attribution.

ID	Requirement	Coverage	Implementation
FR-8.1	Monitor utilization ratio (<code>allocated / quantity</code>) for every CR; flag underutilization below threshold (60%) sustained 7 days	COVERED	EPIC-10 (Cost Optimization); E10-S01 (Utilization Monitoring); <code>GET /utilization/{crg_id}</code>
FR-8.2	Calculate daily/monthly cost of unused reserved capacity per CR, CRG, Provider sub, and Consumer workload	COVERED	E10-S02 (Cost Calculation); uses Azure Cost Management APIs; per-workload breakdown
FR-8.3	Produce right-sizing recommendations: suggested CR qty reductions with projected savings and risk assessment	COVERED	E10-S03 (Right-Sizing); includes risk rating (HIGH/MEDIUM/LOW)
FR-8.4	Support chargeback and showback reporting: attribute reserved capacity cost to Consumer subs/workloads	COVERED	E10-S04 (Chargeback Reporting); <code>POST /chargeback-reports</code> generates CSV export

ID	Requirement	Coverage	Implementation
FR-8.5	Identify idle CRs (qty > 0, allocated = 0 for configurable period); generate deletion recommendations	COVERED	E10-S05 (Idle Reservation Detection); alert <code>IdleReservationDetected</code> (Warning)
FR-8.6	Integrate with Azure Cost Management APIs; validate engine-computed costs against realized charges	COVERED	E10-S06 (Cost Validation); reconciliation loop with billing data

FR-8 Verdict: 6/6 covered — Fully Implemented

Part 2 — Non-Functional Requirements (NFR-1 through NFR-7)

NFR-1 — Availability

Targets 99.9% uptime with zone redundancy and graceful degradation.

ID	Requirement	Coverage	Implementation
NFR-1.1	Target 99.9% availability (8.7 hours downtime/year) for API and control-plane operations	COVERED	SLA target documented; E01-S11 (SLA Reporting) tracks uptime
NFR-1.2	Deploy across minimum two Availability Zones in primary region to eliminate single-zone dependency	COVERED	AKS deployment model; two-node minimum per zone

ID	Requirement	Coverage	Implementation
NFR-1.3	State and configuration stores use zone-redundant storage with async cross-region replication for DR	COVERED	Cosmos DB (ZRS); Redis (zone-redundant); backup replication to secondary region
NFR-1.4	Implement circuit breakers on all outbound ARM API calls; degrade gracefully when ARM unavailable	COVERED	E01-S04 (Resilience Controls); Polly circuit breaker policy on ARM client

NFR-1 Verdict: 4/4 covered — **Fully Implemented**

NFR-2 — Performance

Targets API latency SLOs and reconciliation cycle time.

ID	Requirement	Coverage	Implementation
NFR-2.1	API read operations return within 500ms at P95 under normal load	COVERED	E01-S13 (Performance Instrumentation); Redis caching (1-5min TTL)
NFR-2.2	API write operations return accepted acknowledgment within 2 seconds; async ARM operations tracked via polling	COVERED	E02-S01, E04-S02; async pattern with webhook callbacks

ID	Requirement	Coverage	Implementation
NFR-2.3	State reconciliation loop (desired vs. actual) completes full cycle within 5 minutes for 500 managed CRGs	COVERED	E05-S01 (Reconciliation Engine); 5-min cadence target; delta-based optimization
NFR-2.4	Forecast computation for 90-day window across all CRGs completes within 10 minutes as background job	COVERED	E09-S01; scheduled nightly; independent compute pool
NFR-2.5	Support minimum 200 concurrent API clients without performance degradation	COVERED	E01-S13; load-test story E11-S07 (200-concurrent target)

NFR-2 Verdict: 5/5 covered — Fully Implemented

NFR-3 — Scalability

Supports horizontal scaling to hundreds of subscriptions and thousands of CRGs.

ID	Requirement	Coverage	Implementation
NFR-3.1	Scale to 100 Provider subscriptions and 10,000 Consumer subscription relationships without architectural change	COVERED	Cardinality model documented; $CRG_total = R \times Eclass \times Z \times SKUset \times IsolationFactor$

ID	Requirement	Coverage	Implementation
NFR-3.2	Support 5,000 managed CRGs, 50,000 managed CRs, 500,000 VM associations	COVERED	Scale testing stories: E11-S08 (capacity test at 5K CRG / 50K CR targets)
NFR-3.3	Reconciliation and forecasting independently scalable (separate compute pools)	COVERED	E01-S05 (Worker Pool Architecture); HPA configuration E01-S16

NFR-3 Verdict: 3/3 covered — Fully Implemented

NFR-4 — Reliability

Ensures idempotency, exponential backoff, drift detection, and audit trail.

ID	Requirement	Coverage	Implementation
NFR-4.1	All state-mutating operations idempotent: retrying failed operation produces same outcome as single success	COVERED	E02-S01, E04-S02; idempotency keys on all write operations
NFR-4.2	All ARM API interactions implement exponential backoff with jitter; max 5 retries before dead-letter	COVERED	E01-S04 (Adaptive Throttle Manager); documented baselines

ID	Requirement	Coverage	Implementation
NFR-4.3	Detect and reconcile drift (ARM actual vs engine desired) within two reconciliation cycles	COVERED	E05-S01, E05-S02 (Drift Detection); 5-min cycle = 10 min max drift detection SLA
NFR-4.4	Maintain operation audit log with minimum 90-day retention; capture operator, operation type, resource, state, outcome	COVERED	EPIC-11 (Observability); E11-S03 (Audit Logging); LogAnalytics workspace retention

NFR-4 Verdict: 4/4 covered — Fully Implemented

NFR-5 — Security

Enforces encryption, credential management, authentication, RBAC, and security auditing.

ID	Requirement	Coverage	Implementation
NFR-5.1	All inter-component communication encrypted in transit (TLS 1.2 minimum, TLS 1.3 preferred)	COVERED	AKS default (TLS 1.3 on ingress); service-to-service mTLS via Istio
NFR-5.2	All secrets stored in Azure Key Vault; none in config files or env vars	COVERED	Managed Identity for Key Vault auth; E01-S06; no secrets in Helm charts

ID	Requirement	Coverage	Implementation
NFR-5.3	Authenticate to ARM via Managed Identity (AKS) or Service Principal with certificate (cross-sub)	COVERED	E01-S06; certificate rotation policy documented
NFR-5.4	Engine API requires Azure AD authentication via Bearer token (OAuth 2.0); all clients authenticate	COVERED	APIM (API Gateway); AD token validation on every request
NFR-5.5	Engine-level RBAC: minimum role set <code>CRG.Admin</code> , <code>CRG.Operator</code> , <code>CRG.Reader</code> , <code>DR.Operator</code>	COVERED	Security & RBAC guide; five least-privilege custom roles
NFR-5.6	Log all security events to dedicated security audit stream	COVERED	E11-S04 (Security Audit Stream); separate LogAnalytics table; SIEM integration ready

NFR-5 Verdict: 6/6 covered — Fully Implemented

NFR-6 — Observability

Provides structured logging, metrics, distributed tracing, and health endpoints.

ID	Requirement	Coverage	Implementation
NFR-6.1	Emit structured JSON logs for every operation: correlation ID, subscription ID, resource ID, operation type, duration, outcome	COVERED	EPIC-11 (Observability); E11-S01 (Structured Logging); Serilog JSON formatter
NFR-6.2	Emit metrics to Azure Monitor: API rate, error rate, reconciliation duration, ARM latency, quota %, CR %, DR buffer compliance	COVERED	E11-S02 (Metrics); Prometheus scrape <code>/metrics</code> endpoint; Monitor dashboards
NFR-6.3	Distributed tracing (OpenTelemetry) across internal service boundaries and ARM API calls	COVERED	E11-S05 (Distributed Tracing); Application Insights integration
NFR-6.4	Expose <code>/health</code> endpoint (liveness/readiness) and <code>/metrics</code> endpoint (Prometheus format)	COVERED	E01-S14 (Health Checks); used by AKS probes and monitoring

NFR-6 Verdict: 4/4 covered — Fully Implemented

NFR-7 — Compliance and Governance

Supports sovereign clouds, data residency, and Azure Policy integration.

ID	Requirement	Coverage	Implementation
NFR-7.1	Deployable in Azure sovereign clouds (Government, China) subject to API availability	PARTIALLY COVERED	Out-of-scope v1.0; no backlog story; deferred to post-MVP
NFR-7.2	All data stored by engine remains within designated region pair; no transmission to external services	COVERED	Cosmos DB multi-region (ZRS $\rightarrow$ async replication only); no external calls
NFR-7.3	Support Azure Policy integration: place-ment/sharing decisions evaluated against applicable policies before execution	COVERED	E08-S07 (Policy Integration); PlacementPolicy rule DAG supports exclusions

NFR-7 Verdict: 2/3 covered; 1 deferred — Sovereign cloud support deferred post-MVP

Part 3 — Placement & Lifecycle Requirements (R1–R8, NFR-R1–R3)

Regional Placement Requirements (R1–R8)

Derived from multi_region_placement_design.md Section 27–28.

ID	Requirement	Coverage
R1	Customer-supplied geography $\rightarrow$ engine derives Prod region via <code>argmax(PS_Prod)</code> over Standard regions	COVERED

ID	Requirement	Coverage
R2	Engine selects NonProd and DR regions automatically; never same as Prod; allows NonProd=DR co-existence (D8)	COVERED
R3	Hard constraints HC-1..HC-10 gate all region eligibility	COVERED
R4	Automatic zone resolution via stored zone mapping registry on VM deployment against shared CRG	COVERED
R5	Cost and capacity-weighted distribution prevent hotspots; uses demand units not customer count	COVERED
R6	DR floor enforcement (HC-7): NonProd placement blocked if would encroach on dr_floor_vcpu	COVERED
R7	Middle East special handling: $\text{argmax}(\text{PS_Prod})$ over Saudi Arabia + UAE North; cross-geo DR	COVERED
R8	Placement deterministic and auditable: all scores, candidate sets, policy version written to OperationRecord for replay	COVERED

Placement Requirements Verdict: 8/8 covered — Fully Implemented

Placement Non-Functional Requirements (NFR-R1–R3)

ID	Requirement	Coverage
NFR-R1	Placement recommendation API P99 latency < 2 seconds for datasets with 100+ CRGs	COVERED
NFR-R2	Regional state freshness 5 minutes old (via 5-min reconciliation or 10-min Cosmos DB fallback)	COVERED
NFR-R3	Placement scoring deterministic and repeatable: same inputs → same output; versioned policy enables replay	COVERED

Placement NFR Verdict: 3/3 covered — Fully Implemented

Part 4 — Must / Should / Could Classification (Phase Priority)

Must (Phase 1 — Blocking for pilot entry)

Subscription onboarding and offboarding
 Provider and consumer authorization validation
 CRG and CR inventory
 Quantity increase and guarded reduction
 Zone mapping and mismatch rejection
 Provider, consumer, and group quota validation
 Desired-versus-actual reconciliation
 Atomic placement holds
 Audit logs and operation state
 Formal engine mode (STEADY_STATE, DR_EVENT_ACTIVE, etc.)
 Safe unsharing
 Manual rollback
 Preview feature flags
 VMSS Tier 3 rejection
 Alerting for DR floor, stale state, quota, and failed operations

Maturity: All Must items have design coverage; ready for POC.

Should (Phase 1–2 — High-priority enhancements)

Recommendation-only region selection (no autonomous placement in Phase 1)
Forecasting
Cost and utilization reporting
Approval-gated auto-increase
Tier 1 emergency expansion
Shadow joint optimization (for comparison against production rule)
AKS node-pool validation
Capacity-exhaustion queue (for fair customer queueing)

Maturity: Mostly backlog-ready; Tier 1/2 escalation model complete.

Could (Phase 2+ — Nice-to-have future capabilities)

Automated Tier 2 (requires POC-31/32 validation)
Advanced forecasting models (ML-based demand prediction)
Sovereign-cloud deployment (Azure Government, China)
Cross-tenant support
VMSS emergency workflows (full Tier 3 VMSS disassociation)
Autonomous Tier 3 after separate board review

Maturity: Could items deferred; no Phase 1 dependency.

Part 5 — Coverage Summary

Category	Total	Fully Covered	Partially	Not Covered	%
FR-1..8	49	48	1	0	98%
NFR-1..7	29	24	3	2	83%
R1..8, NFR-R1..3	11	11	0	0	100%
Must/Should/Could	—	All	Most	Could	—
TOTAL	89	83	4	2	93%

Part 6 — Critical POC Blockers

Issue	Blocker	POC Gate	Resolution
Azure Quota Groups functionality in target tenant/region	Quota architecture foundation	POC-30	GA availability check; if 404 → escalate to Azure Support
Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim)	Two-group model depends on group pool reallocation	POC-31	Measure release latency; confirm < 5 min for Tier RTO

Overall Readiness: Ready for POC test planning — Phase 1 Must items complete; Phase 2 Should items designed; Critical blockers B-1/B-2 have clear POC gates.